

“AI-Powered Humanoid Robot Fleet Management System”

Team: Robo-Nexus

Team Members:

Kiran Khatri	kkhatri@sfsu.edu
Kerry Yu	kyu19@sfsu.edu
Dias Almat	dalmat@sfsu.edu
Ishaank Zalpuri	izalpuri@sfsu.edu
Etienne Ghashehbaba	eghashehbaba@mail.sfsu.edu
Shadi Daher	sdaher@sfsu.edu

GitHub Repository:

<https://github.com/sfsu-joseo/csc675-775-database-api-project-robo-nexus>

Version	Description
V2	Final Submission

Table of contents

1. [Project Description](#)
2. [Functional Database Requirements](#)
3. [Non-Functional Database Requirements](#)
4. [List of Main Data Items and Entities](#)

Project Description

This project focuses on designing a database system to manage fleets of humanoid robots used in environments such as warehouses, hospitals, and homes. As humanoid robots become more common, they generate a large amount of information, including task assignments, task execution history, deployment locations, energy levels, maintenance records, and interaction details. Currently, much of this information is either spread across different tools or not stored in a structured way, making it difficult to track robot behavior and performance over time.

There are existing robot fleet management platforms, such as `Formant` and `InOrbit`, that help monitor robots and show their current status. These systems are useful for real-time monitoring and coordination, but they are mainly designed for general robot fleets and do not focus on organizing long-term data for humanoid robots. In particular, they do not emphasize storing detailed interaction context, coordination roles, or historical data in a centralized database.

One unique feature of the system is its ability to store *human emotional context data* in personal and healthcare environments. In settings such as hospitals or homes, humanoid robots often interact closely with people. The system allows simple emotional observations, such as calm or urgent situations, to be recorded as data for later analysis. Most existing fleet management tools do not capture or organize this type of information.

Another unique feature is *role-based robot coordination* in warehouse environments. The system allows certain robots to be assigned a *senior* role to help coordinate tasks among other robots based on factors such as hardware capability and energy level. If a senior robot becomes unavailable, the system supports reassigning that role to another robot, helping prevent disruptions. Existing tools typically treat all robots equally and do not support this type of structured coordination with a fail-safe mechanism.

Overall, this project provides a centralized and structured way to manage humanoid robot data that existing solutions do not fully address. Manufacturers, warehouses, hospitals, and households deploying humanoid robots can benefit from a system that keeps robot information organized, reliable, and easy to understand over time.

Functional Database Requirements

1. Robot (Strong)

- 1.1. A robot shall belong at most one robot model.
- 1.2. A robot shall be deployed at at most one deployment location at a time.
- 1.3. A robot shall be assigned at most one role at a time.
- 1.4. A robot shall have many task assignments.
- 1.5. A robot shall participate in many task executions.
- 1.6. A robot shall have many emotion records over time.
- 1.7. A robot shall have at most one user.
- 1.8. A robot shall have many energy status records over time.
- 1.9. A robot shall generate many logs.
- 1.10. A robot may be associated with many RobotHardware.
- 1.11. A robot shall have at most one software.
- 1.12. A robot shall have many performance metrics.
- 1.13. A robot shall generate many alerts.
- 1.14. A robot shall have many diagnostic reports over time.
- 1.15. A robot shall have many sensors
- 1.16. A robot shall charge at many stations
- 1.17. A robot shall have many charging sessions
- 1.18. A robot shall have a current status
- 1.19. A robot shall have many maintenance record

2. Robot Model (Strong)

- 2.1. A robot model shall support zero or many hardware components.
- 2.2. A robot model shall be produced by at most one manufacturer.
- 2.3. A robot model shall be associated with many robots.

3. Operator (User) (Strong)

- 3.1. An operator shall interact with many robots.
- 3.2. An operator may participate in many task assignments.
- 3.3. An operator shall assign many robot roles.
- 3.4. An operator may participate in many interaction sessions.

3.5. An operator shall be assigned many deployment locations.

4. Manufacturer (Strong)

- 4.1. A manufacturer shall produce many robot models.
- 4.2. A manufacturer shall produce many hardware.
- 4.3. A manufacturer shall provide zero or many software versions.

5. Hardware(Strong)

- 5.1. A hardware component shall be produced by exactly one manufacturer.
- 5.2. A hardware component shall have many maintenance records.
- 5.3. A hardware component shall be supported by many robot models.

6. Robot Hardware(Weak)

- 6.1. A robot hardware record shall belong to exactly one robot.
- 6.2. A robot hardware record shall belong to exactly one hardware component.
- 6.3. A robot hardware shall be assigned by at least one technician.

7. Software Version (Strong)

- 7.1. A software version shall be associated with many robots
- 7.2. A software version shall be provided by exactly one manufacturer.
- 7.3. A software version shall be updated by at most one technician.

8. Deployment Location (Strong-Superclass)

- 8.1. A deployment location shall host many robots.
- 8.2. A deployment location shall support different environment types (warehouse, hospital, household).
- 8.3. A deployment location shall belong to at most one address.
- 8.4. A deployment location shall host many users
- 8.5. A deployment location shall have many charging sessions

9. Address (Strong)

- 9.1. An address may be associated with many deployment locations.
- 9.2. An address shall store street, city, state, and zipcode information.
- 9.3. An address shall be associated with many users.

10. Task (Strong)

- 10.1. A task shall be associated with many task assignments.
- 10.2. A task shall be associated with many task executions.
- 10.3. A task shall be associated with at most one AI response.

11. Task Assignment (weak)

- 11.1. A task assignment shall associate exactly one robot .
- 11.2. A task assignment shall belong to exactly one operator.
- 11.3. A task assignment shall belong to exactly one task.

12. Task Execution (Weak)

- 12.1. A task execution shall belong to exactly one robot.
- 12.2. A task execution shall belong to exactly one task.
- 12.3. A task execution shall generate at least one log

13. Emotion Record (Weak)

- 13.1. An emotion record shall belong to exactly one interaction session.
- 13.2. An emotion record shall belong to at least one hospital.
- 13.3. An emotion record shall have at least one AI request.
- 13.4. An emotion record shall be summarized by one emotional pattern

14. InteractionSession(Weak)

- 14.1. An interaction session shall belong to exactly one robot.
- 14.2. An interaction session shall belong to exactly one operator.
- 14.3. An interaction session shall belong to exactly one home.

15. Robot Role (Strong)

- 15.1. A robot role shall be assigned to many robots.
- 15.2. A robot role shall be assigned by at most one user.
- 15.3. A robot role may be recommended by at most one AI response.

16. Energy Status (Weak)

- 16.1. An energy status shall be associated with exactly one robot.
- 16.2. An energy status shall be linked to one charging session.
- 16.3. An energy status shall have at least one logs.

17. PerformanceMetric(Weak)

- 17.1. A performance metrics shall belong to exactly one robot.
- 17.2. A performance shall have a time stamp record.
- 17.3. A performance metric shall be associated with at least one diagnostic report.

18. Alert(Weak)

- 18.1. An alert shall belong to exactly one robot.
- 18.2. An alert shall record severity and time.
- 18.3. An alert may be associated with one AI request.

19. DiagnosticReport(Weak)

- 19.1. A diagnostic report shall belong to exactly one robot.
- 19.2. A diagnostic report shall have performance metrics.
- 19.3. A diagnostic report shall be associated with one AI model.

20. Log (weak)

- 20.1. A log shall be associated with one robot.
- 20.2. A log shall be generated by at least one sensor.
- 20.3. A log shall have at least one charging session information.

21. Sensor (Weak)

- 21.1. A sensor shall be associated with one robot.
- 21.2. A sensor shall generate at least one log.
- 21.3. A sensor shall belong to at least one manufacturer.

22. Charging Station (Strong)

- 22.1. A charging station shall serve many robots.
- 22.2. A charging station shall record many charging sessions.
- 22.3. A charging station shall have many maintenance records.

23. Charging Session (Weak)

- 23.1. A charging session shall occur at one charging station.
- 23.2. A charging session shall be associated with one robot.
- 23.3. A charging session shall be associated with one deployment location.
- 23.4. A charging session shall generate at least one log.

24. Support Request (Weak)

- 24.1. A support request shall belong to exactly one robot.
- 24.2. A support request shall belong to exactly one technician.
- 24.3. A support request shall be associated with at least one maintenance record.

25. AI Model (Strong)

- 25.1. An AI model shall generate many AI requests.
- 25.2. An AI model shall analyze zero or many diagnostic reports.
- 25.3. An AI model shall analyze many emotional patterns.

26. AI Request(Weak)

- 26.1. An AI request shall belong to exactly one AI model.
- 26.2. An AI request shall reference one robot.
- 26.3. An AI request shall have at least one emotion record.
- 26.4. An AI request shall belong to one AI response.

27. AI Response (Weak)

- 27.1. An AI response shall belong to exactly one AI request.
- 27.2. An AI response shall recommend exactly one robot role.
- 27.3. An AI response shall recommend one task.

28. Hospital (Weak-Subclass)

- 28.1. A hospital shall belong to at least one deployment location.
- 28.2. A hospital shall have at least one department.
- 28.3. A hospital shall have at least one emotion record.

29. Warehouse (Weak-Subclass)

- 29.1. A warehouse shall belong to at least one deployment location.
- 29.2. A warehouse shall host at least one robot.
- 29.3. A warehouse shall have at least one storage zone.

30. Home (Weak-Subclass)

- 30.1. A home shall belong to at least one deployment location.
- 30.2. A home may host at least one robot.
- 30.3. A home shall have at least one interaction session.

31. Factory (Weak-Subclass)

- 31.1. A factory shall belong to at most one deployment location.
- 31.2. A factory shall have at least one robot.
- 31.3. A factory shall have at least one product.

32. Department(Weak)

- 32.1. A department shall belong to exactly one hospital.
- 32.2. A department may host at least one robot.
- 32.3. A department may record at least one interaction session.

33. EmotionalPattern (Weak)

- 33.1. An emotional pattern shall belong to exactly one robot.
- 33.2. An emotional pattern shall summarize at least one emotion record.
- 33.3. An emotional pattern shall be analyzed by exactly one AI model.

34. MaintenanceRecord (Weak)

- 34.1. A maintenance record shall belong to exactly one robot.
- 34.2. A maintenance record shall reference at least one hardware component.
- 34.3. A maintenance record shall be performed by exactly one technician.

35. Technician

- 35.1. A technician shall perform zero or many maintenance records.
- 35.2. A technician shall resolve zero or many support requests.
- 35.3. A technician shall be assigned to at most one deployment location.

Non-Functional Database Requirements

1. Performance

- 1.1. The system shall support concurrent access by at least 1000 robots without performance degradation.
- 1.2. The system shall be able to handle concurrent usage by 100 robots
- 1.3. The system shall be able to maintain performance while receiving AI driven queries for real time data

2. Security

- 2.1. Any logs and data shall only be able to be accessed by a user with sufficient permissions
- 2.2. The system shall protect AI request and AI response data from unauthorized access.
- 2.3. The system shall protect sensitive data to be modified by robots.

3. Reliability

- 3.1. The system shall retain operational logs for at least 12 months.
- 3.2. The system shall have daily backups
- 3.3. The system shall continue operating if there is an issue

4. Maintainability and Coding Standards

- 4.1. The system shall host AI queries and structural queries separately
- 4.2. The database schema shall be well documented.
- 4.3. SQL keywords shall be written in upper case

5. Scalability

- 5.1. The system shall support growth in the number of robots and deployment locations.
- 5.2. The system shall support increasing volumes of task execution and logs data.
- 5.3. The system shall support any future AI integration.

6. Privacy

- 6.1. Access to emotional interaction records shall be restricted.
- 6.2. Access to emotional data shall be restricted to authorized persons only.
- 6.3. Access to operator information from the home subclass shall be restricted.

7. Storage

- 7.1. Emotional interaction records shall be stored securely to protect human-related sensitive data.
- 7.2. The system shall store logs, task history, charging session history.
- 7.3. The system shall store high volumes of data related to performance metrics and AI responses.

List of Main Data Items and Entities

1. **Robot:** Strong

- 1.1 robot_id: key, numeric (PK)
- 1.2 charging_station_id (FK)
- 1.3 robotmodel_id: numeric (FK)
- 1.4 deployment_location_id: numeric (FK)
- 1.5 alert_id: numeric (FK)
- 1.6 user_id: numeric (FK, optional)
- 1.7 software_version_id: numeric (FK)
- 1.8 manufacturer_id: numeric (FK)
- 1.9 performance_metric_id: numeric(FK)

2. **RobotModel:** Strong

- 2.1 robotmodel_id: key, numeric (PK)
- 2.2 manufacturer_id: numeric (FK)
- 2.3 model: enum, alphanumeric

3. **User:** Strong

- 3.1. user_id: key, numeric (PK)
- 3.2. user_name: key, alphanumeric

4. **Manufacturer:** Strong
 - 4.1 manufacturer_id: key, numeric (PK)
 - 4.2 name: enum, alphanumeric
5. **Hardware:** Strong
 - 5.1 hardware_id: key, numeric (PK)
 - 5.2 manufacturer_id: numeric (FK)
6. **RobotHardware:** Weak
 - 6.1 robot_hardware_id: key, numeric
 - 6.2 robot_id: numeric (FK)
7. **SoftwareVersion:** Strong
 - 7.1 software_version_id: key, numeric
8. **DeploymentLocation:** Strong (Superclass)
 - 8.1 deployment_location_id: key, numeric (PK)
 - 8.2 technician_id: numeric(FK)
 - 8.3 charging_session_id: numeric (FK)
 - 8.4 environment_type: enum, alphanumeric
(warehouse/hospital/home/factory)
9. **Address:** Strong
 - 9.1 address_id: key, numeric (PK)
 - 9.2 street: alphanumeric
 - 9.3 city: alphanumeric
 - 9.4 state: alphanumeric
 - 9.5 zipcode: alphanumeric
10. **Task:** Strong
 - 10.1 task_id: key, numeric (PK)
 - 10.1 ai_response_id: numeric (FK)
 - 10.2 task_name: alphanumeric
 - 10.3 description: text
 - 10.4 priority: enum, numeric
 - 10.5 status: bool (Complete)

- 11. **TaskAssignment:** Weak
 - 11.1 task_assignment_id: key, numeric (PK)
 - 11.2 robot_id: numeric (FK)
 - 11.3 user_id: numeric (FK)
 - 11.4 task_id: numeric (FK)
 - 11.5 assigned_at: timestamp
- 12. **TaskExecution:** Weak
 - 12.1 task_execution_id: key, numeric (PK)
 - 12.3 task_id: numeric (FK)
 - 12.4 start_time: timestamp
 - 12.5 end_time: timestamp
 - 12.6 status: bool, (Successful)
- 13. **EmotionalRecord:** Weak
 - 13.1 emotional_id: key, numeric (PK)
 - 13.2 pattern_id: numeric (FK)
 - 13.3 emotional_state: enum, alphanumeric
 - 13.4 timestamp: timestamp
 - 13.5 description: text
- 14. **InteractionSession:** Weak
 - 14.1 interaction_id: key, numeric (PK)
 - 14.2 robot_id: numeric (FK)
 - 14.3 user_id: numeric (FK)
 - 14.4 home_id: numeric (FK)
 - 14.4 start_time: timestamp
 - 14.5 end_time: timestamp
- 15. **RobotRole:** Strong
 - 15.1 robotrole_id: key, numeric (PK)
 - 15.2 ai_response_id (FK)
 - 15.3 robotrole: enum, alphanumeric
- 16. **EnergyStatus:** Weak
 - 16.1 energy_status_id: key, numeric (PK)

- 16.2 robot_id: numeric (FK)
- 16.3 charging_session_id: numeric (FK)
- 16.4 battery_level: numeric
- 16.5 timestamp: timestamp
- 17. **PerformanceMetric:** Weak
 - 17.1 performance_metric_id: key, numeric (PK)
 - 17.2 robot_id: numeric (FK)
 - 17.3 metric_type: alphanumeric
 - 17.4 metric_value: numeric
 - 17.5 timestamp: timestamp
- 18. **Alert:** Weak
 - 18.1 alert_id: key, numeric (PK)
 - 18.2 robot_id: numeric (FK)
 - 18.3 severity: enum, alphanumeric
 - 18.4 timestamp: timestamp
- 19. **DiagnosticReport:** Weak
 - 19.1 diagnostic_report_id: key, numeric (PK)
 - 19.2 robot_id: numeric (FK)
 - 19.3 ai_model_id: numeric (FK)
 - 19.4 generated_at: timestamp
 - 19.5 summary: text
- 20. **Log:** Weak
 - 20.1 log_id: key, numeric (PK)
 - 20.2 robot_id: numeric (FK)
 - 20.3 deployment_location_id: numeric (FK, optional)
 - 20.4 timestamp: timestamp
 - 20.5 event_type: alphanumeric
 - 20.6 event_details: text
- 21. **Sensor:** Weak
 - 21.1 sensor_id: key, numeric (PK)
 - 21.2 robot_id: numeric (FK)

- 21.3 sensor_type: alphanumeric
- 21.4 status: enum, alphanumeric
- 22. **ChargingStation:** Strong
 - 22.1 charging_station_id: key, numeric (PK)
 - 22.2 station_name: alphanumeric
 - 22.3 capacity: numeric
 - 22.4 status: enum, alphanumeric
- 23. **ChargingSession:** Weak
 - 23.1 charging_session_id: key, numeric (PK)
 - 23.2 charging_station_id: numeric (FK)
 - 23.3 robot_id: numeric (FK)
 - 23.4 deployment_location_id: numeric (FK)
 - 23.5 start_time: timestamp
 - 23.6 end_time: timestamp
- 24. **SupportRequest:** Weak
 - 24.1 support_request_id: key, numeric (PK)
 - 24.2 robot_id: numeric (FK)
 - 24.3 user_id: numeric (FK)
 - 24.4 issue_details: text
 - 24.5 time_reported: timestamp
- 25. **AIModel:** Strong
 - 25.1 aimodel_id: key, numeric (PK)
 - 25.2 model_name: alphanumeric
 - 25.3 service_type: alphanumeric
 - 25.4 version_label: alphanumeric
- 26. **AIRequest:** Weak
 - 26.1 ai_request_id: key, numeric (PK)
 - 26.2 aimodel_id: numeric (FK)
 - 26.3 robot_id: numeric (FK)
 - 26.4 requested_at: timestamp

- 27. **AIResponse:** Weak
 - 27.1 ai_response_id: key, numeric (PK)
 - 27.2 ai_request_id: numeric (FK)
 - 27.3 recommended_role_id: numeric (FK)
 - 27.4 recommended_task_id: numeric (FK)
 - 27.5 response_payload: text
 - 27.6 generated_at: timestamp
- 28. **Hospital:** Weak (Subclass)
 - 28.1 hospital_id: key, numeric (PK)
 - 28.2 deployment_location_id: numeric (FK, unique)
- 29. **Warehouse:** Weak (Subclass)
 - 29.1 warehouse_id: key, numeric (PK)
 - 29.2 deployment_location_id: numeric (FK, unique)
- 30. **Home:** Weak (Subclass)
 - 30.1 home_id: key, numeric (PK)
 - 30.2 deployment_location_id: numeric (FK, unique)
- 31. **Factory:** Weak (Subclass)
 - 31.1 factory_id: key, numeric (PK)
 - 31.2 deployment_location_id: numeric (FK, unique)
- 32. **Department:** Weak
 - 32.1 department_id: key, numeric (PK)
 - 32.2 hospital_id: numeric (FK)
 - 32.3 department_name: alphanumeric
- 33. **EmotionalPattern:** Weak
 - 33.1 emotional_pattern_id: key, numeric (PK)
 - 33.2 robot_id: numeric (FK)
 - 33.3 ai_model_id: numeric (FK)
 - 33.4 pattern_summary: text
- 34. **MaintenanceRecord:** Weak
 - 34.1 maintenance_record_id: key, numeric (PK)

- 34.2 robot_id: numeric (FK)
 - 34.3 technician_id: numeric (FK)
- 35. **Technician**: Strong
 - 35.1 technician_id: key, numeric (PK)
 - 35.2 name: alphanumeric
- 36. **User_Deployment_Location**: Associative
 - 36.1 charging_station_id: numeric(FK)
 - 36.2 robot_id: numeric(FK)
- 37. **EmotionalAnalyze**: Associative
 - 37.1 emotional_id: numeric(FK)
 - 37.2 ai_request_id: numeric(FK)
- 38. **AIRequestRecord**: Associative
 - 38.1 emotional_id: numeric(FK)
 - 38.2 ai_request_id: numeric(FK)
- 39. **PerformanceDiagnostic**: Associative
 - 39.1 diagnostic_report_id: numeric(FK)
 - 39.2 performance_metric_id: numeric(FK)
- 40. **EmotionalInteractionSession**: Associative
 - 40.1 emotional_id: numeric(FK)
 - 40.2 interaction_id: numeric(FK)
- 41. **Technician_Support_Request**: Associative
 - 41.1 support_request_id: numeric(FK)
 - 41.2 technician_id: numeric(FK)
- 42. **Robot_ChargingStation**: Associative
 - 42.1 charging_station_id: numeric(FK)
 - 42.2 robot_id: numeric(FK)
- 43. **logSession**: Associative
 - 43.1 charging_session_id: numeric(FK)
 - 43.2 log_id: numeric(FK)

- 44. **HardwareMaintenance**: Associative
 - 44.1 hardware_id (FK)
 - 44.2 maintainence_record_id (FK)
 - 45. **ChargingStationSession**: Associative
 - 45.1 charging_session_id: numeric(FK)
 - 45.2 charging_station_id: numeric(FK)
 - 46. **PatientEmotionalRecord**: Associative
 - 46.1 emotional_id: numeric(FK)
 - 46.2 hospital_id: numeric(FK)
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ERD DIAGRAM

